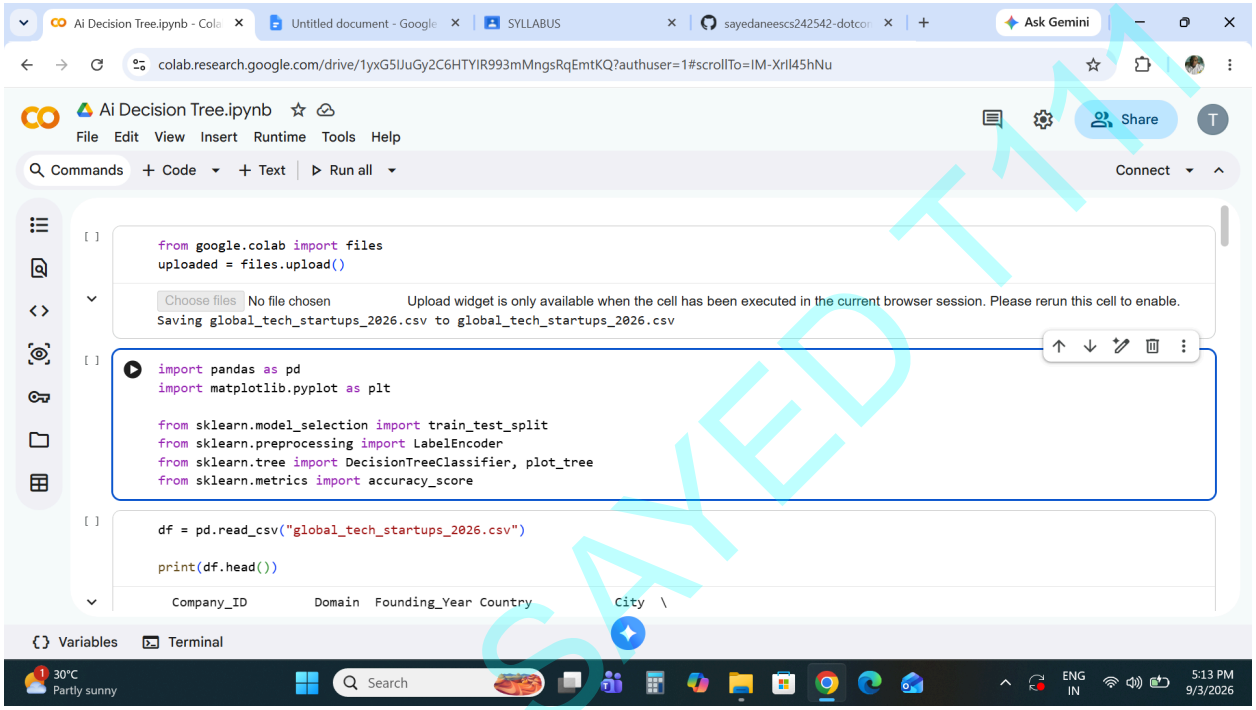
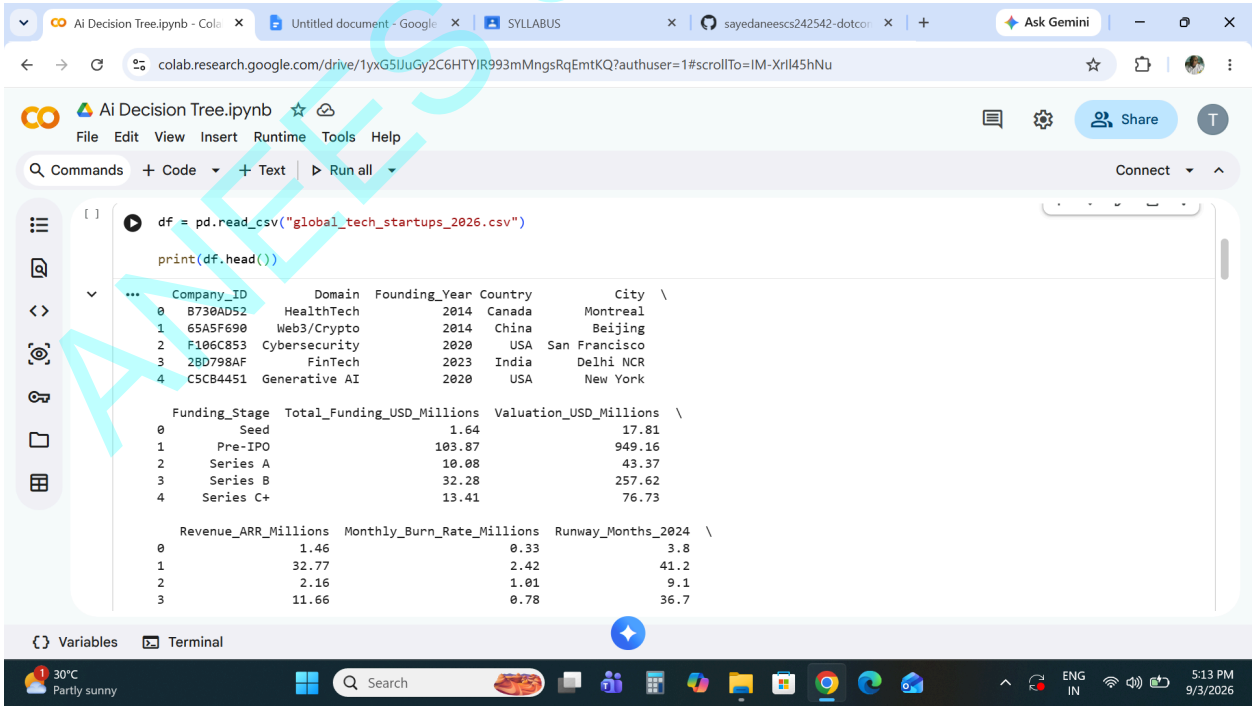
SHETH MVLU COLLEGE
SUBJECT:- AI

Practical 4

AIM:- Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.





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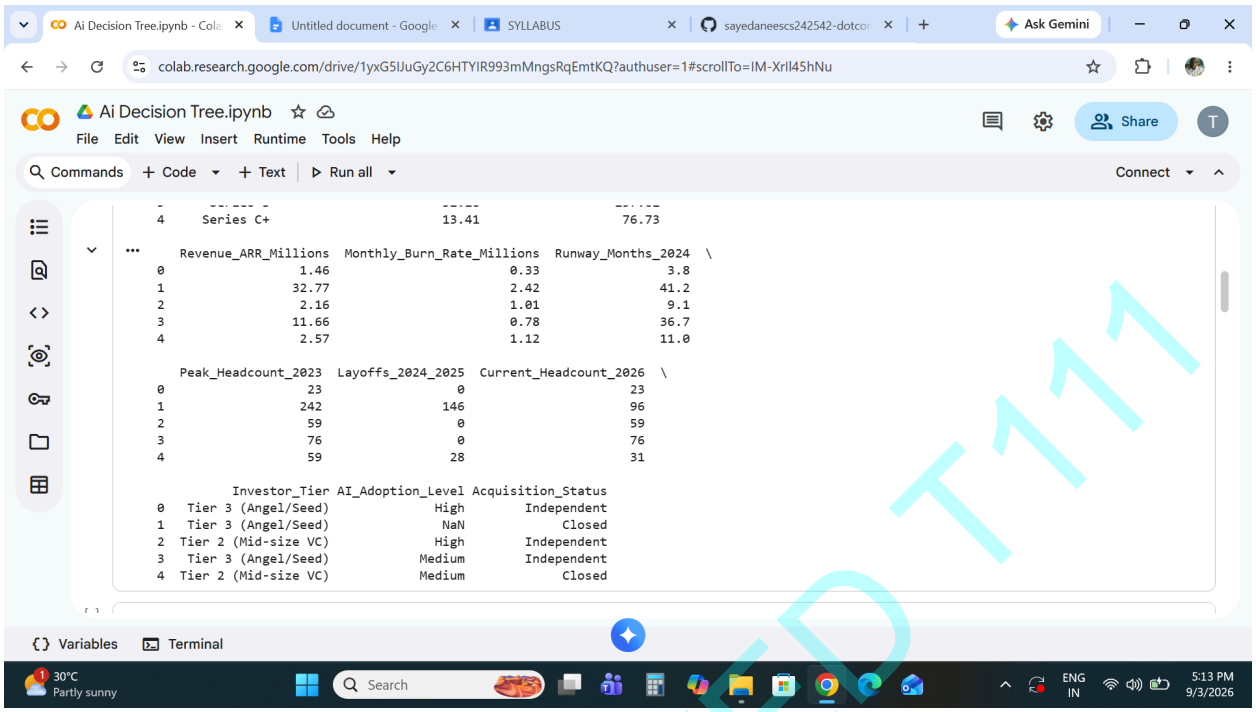
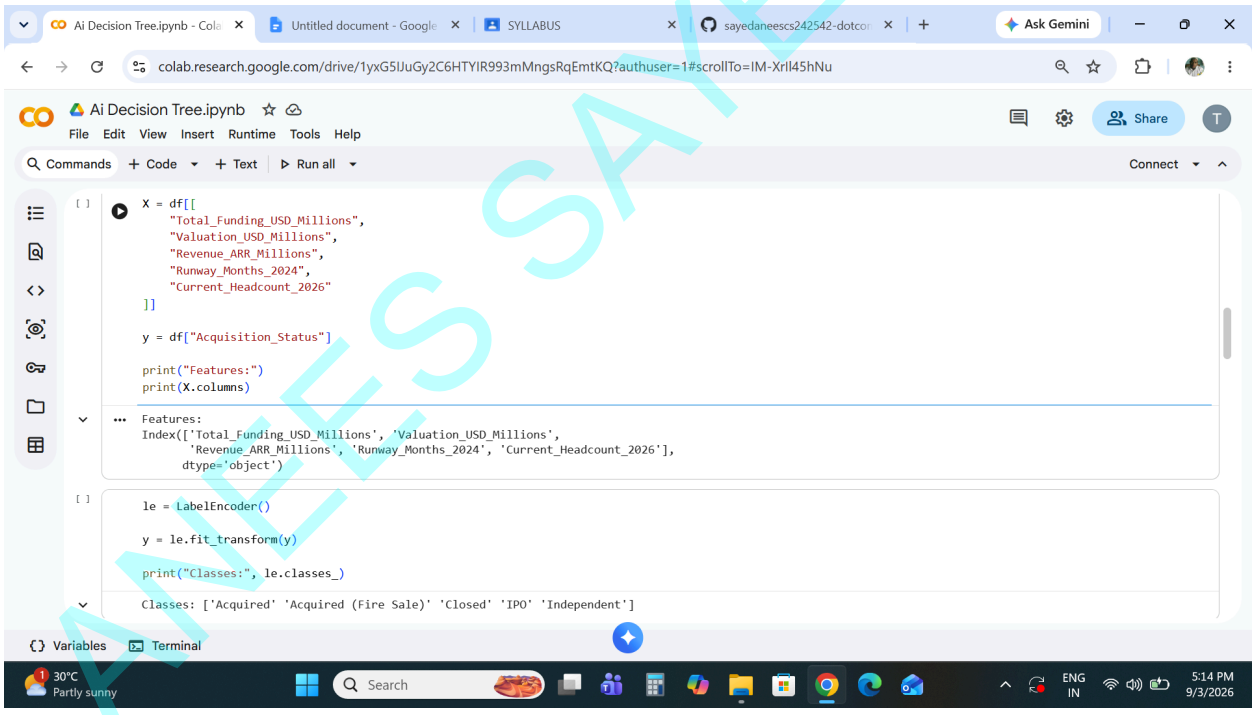


```
4    Series C+                    13.41                 76.73

   Revenue_ARR_Millions  Monthly_Burn_Rate_Millions  Runway_Months_2024  \
0                  1.46                        0.33                 3.8
1                 32.77                        2.42                41.2
2                  2.16                        1.01                 9.1
3                 11.66                        0.78                36.7
4                  2.57                        1.12                11.0

   Peak_Headcount_2023  Layoffs_2024_2025  Current_Headcount_2026  \
0                   23                  0                      23
1                  242                146                      96
2                   59                  0                      59
3                   76                  0                      76
4                   59                 28                      31

          Investor_Tier AI_Adoption_Level Acquisition_Status
0   Tier 3 (Angel/Seed)              High        Independent
1   Tier 3 (Angel/Seed)               NaN             Closed
2  Tier 2 (Mid-size VC)              High        Independent
3   Tier 3 (Angel/Seed)            Medium        Independent
4  Tier 2 (Mid-size VC)            Medium             Closed
```



```
X = df[[
    "Total_Funding_USD_Millions",
    "Valuation_USD_Millions",
    "Revenue_ARR_Millions",
    "Runway_Months_2024",
    "Current_Headcount_2026"
]]

y = df["Acquisition_Status"]

print("Features:")
print(X.columns)
```

```
Features:
Index(['Total_Funding_USD_Millions', 'Valuation_USD_Millions',
       'Revenue_ARR_Millions', 'Runway_Months_2024', 'Current_Headcount_2026'],
      dtype='object')
```

```
le = LabelEncoder()

y = le.fit_transform(y)

print("Classes:", le.classes_)
```

```
Classes: ['Acquired' 'Acquired (Fire Sale)' 'Closed' 'IPO' 'Independent']
```

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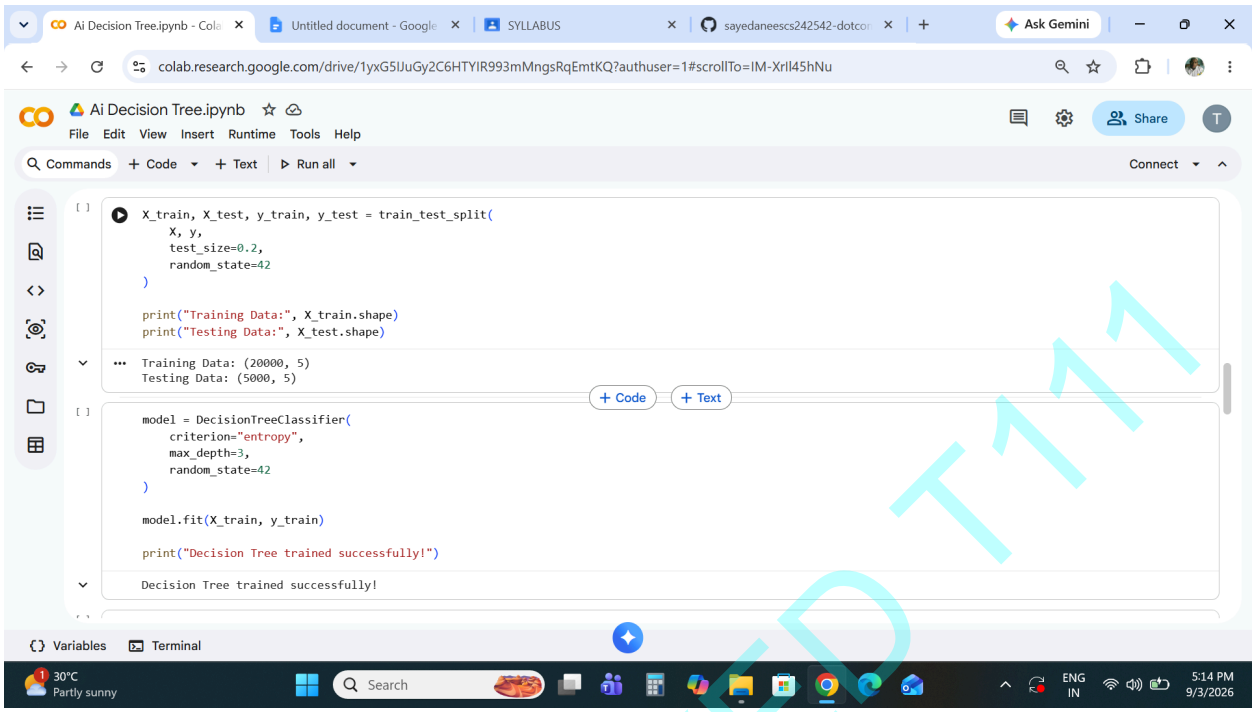

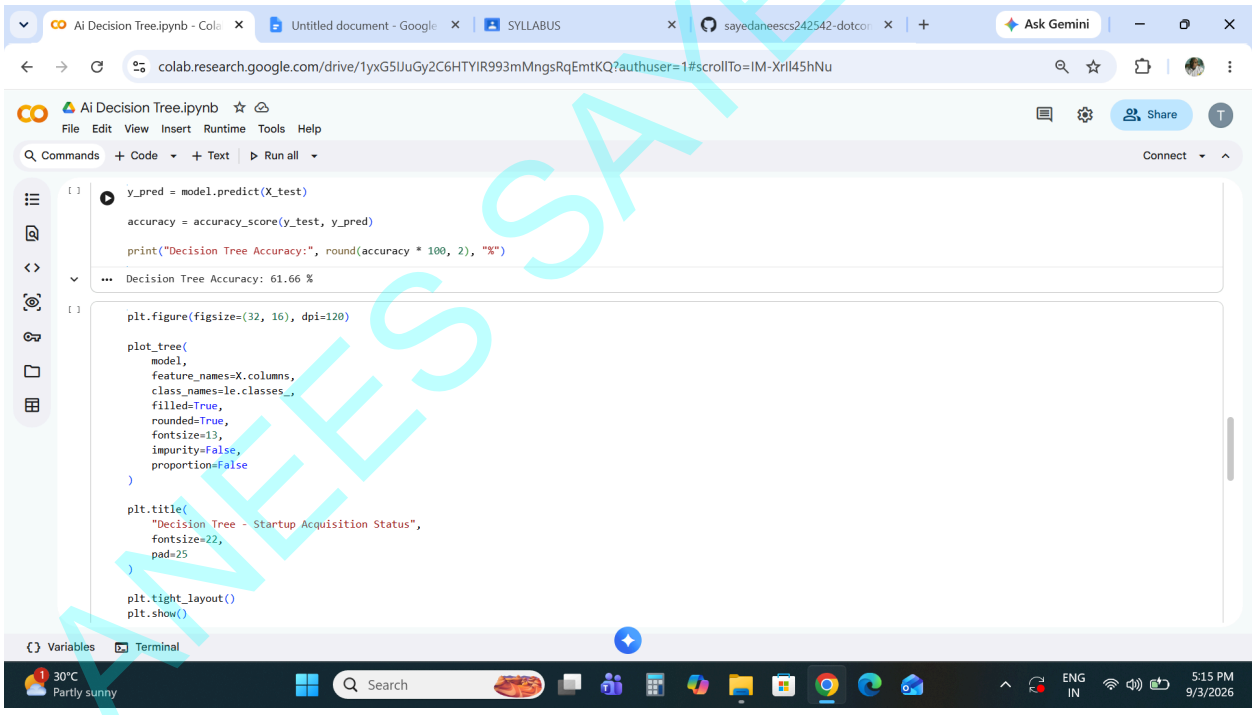

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Decision Tree - Startup Acquisition Status